

Show all work. Indicate if values determined via shock table or code, if not otherwise obvious from the work. Submit any code used. If an LLM is used, attach a separate chat thread for each problem.

ME 561 students are required to typeset assignments using $\text{\LaTeX}$. **ME 461** students receive a 2% bonus for typesetting assignments using $\text{\LaTeX}$.

1. (15 pts) Air enters a convergent-divergent nozzle at $20 \times 10^5 \text{ N/m}^2$ and 40°C . The receiver pressure is $2 \times 10^5 \text{ N/m}^2$ and the nozzle throat area is 10 cm^2 .
 - (a) What should the exit area be for the design conditions above (i.e., to operate at third critical?)
 - (b) With the nozzle area fixed at the value determined in part (a) and the inlet pressure held at $20 \times 10^5 \text{ N/m}^2$, what receiver pressure would cause a shock to stand at the exit?
 - (c) What receiver pressure would place the shock at the throat?
2. (20 pts) A converging-diverging nozzle receives air from a tank at 10 bar and 300 K. The pressure is 2.8 bar immediately preceding a plane shock that is located in the diverging section. The Mach number at the exit is 0.5 and the flow rate is 5 kg/s . Determine:
 - (a) The throat area
 - (b) The area at which the shock is located
 - (c) The outlet pressure required to operate the nozzle in the manner described above.
 - (d) The outlet area
 - (e) The design Mach number
3. (15 pts) Steam is to be expanded to a Mach 2.0 in a converging-diverging nozzle from an inlet velocity of 100 m/s . The inlet area is 50 cm^2 , and the inlet static temperature is 500 K . Assuming isentropic flow, determine the throat and exit areas required. Assume the steam to behave as a perfect gas with constant $\gamma = 1.3$.
4. **ME 561 ONLY** (10 pts) In Figure 1, $M_1 = 3.0$ and $A_1 = 0.2 \text{ m}^2$. If the fluid is carbon monoxide and the shock occurs at an area of 0.15 m^2 , what is the minimum area possible for section 4?

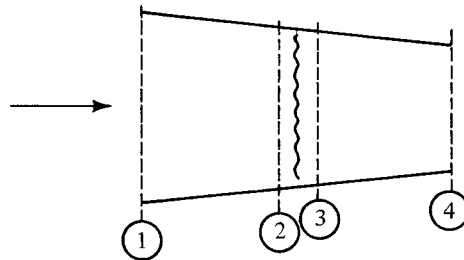
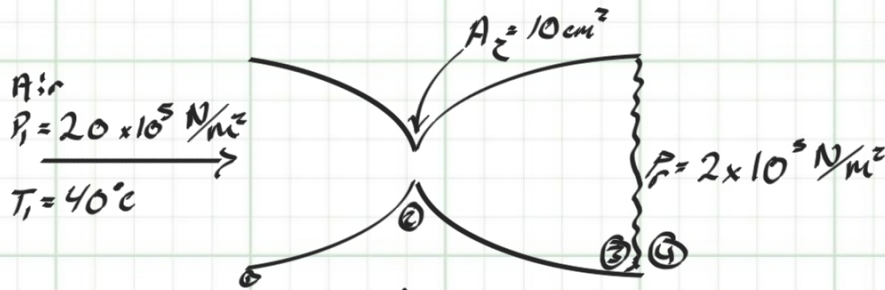


Figure 1: Shock in carbon monoxide flow

Problem 1



Converging diverging Nozzle

- What is the exit Area for this design to operate @ 3rd Critical.
- W/ Nozzle Area fixed to that found in part a) What receiver Pressure would induce a shock @ the exit.
- What P_r would induce shock @ throat?

Known:

$$\frac{P}{P_1} = e^{-\gamma s/R} \quad \frac{P}{P_1} = \frac{A_1^*}{A_2^*} \Rightarrow P_2 A_2^* = P_1 A_1^* \Rightarrow \frac{A_2^*}{A_1^*} = e^{\gamma s/R} \text{ for 1-D flow perfect gas when } Q = W_s = 0$$

$$\therefore \text{isentropic} \quad P_1 = P_{t1} = P_{t2} = P_{t3} = P_{t4} \quad @ \text{ 3rd crit } P_4 = P_r$$

$$\Rightarrow A^* = A_{1-3} \quad \frac{P_{t2}}{P_{t1}} = \frac{A_1^*}{A_2^*} \quad \frac{P_{t3}}{P_{t2}} = \frac{A_2^*}{A_3^*} \quad \frac{P_{t4}}{P_{t3}} = \frac{A_3^*}{A_4^*}$$

$$A_2^* = A_2 = 10 \text{ cm}^2$$

$$\therefore P_4 = P_r \rightarrow P_{t3} = P_1$$

$$\Rightarrow \frac{P_4}{P_{t4}} = \frac{P}{P_{t4}} = 0.1 \Rightarrow M = 2.15 \Rightarrow \frac{A_4}{A^*} = 1.91854$$

$$\Rightarrow A_4 = (1.91854)(10 \text{ cm}^2)$$

Using Appendix G. - isentropic flow Parameters

$$A_4 = 1.92 \text{ cm}^2$$

M	P/P _t	T/T _t	A/A*	PA/P _t A*
2.15	0.101	0.91962	1.91854	0.19403

M ₁	M ₂	P ₂ /P ₁	T ₂ /T ₁	ΔV/a ₁	P ₀₂ /P ₀₁	P ₀₂ /P ₁
2.15	0.55395	5.27625	1.81522	1.40407	0.65105	6.45766
M	P/P _t	T/T _t	A/A*	PA/P _t A*		
0.55	0.81417	0.94295	1.25495	1.02175		

$$P_4 = \frac{P_4}{P_{t4}} \frac{P_{t4}}{P_{t3}} P_1 \quad P_4 = P_r = (0.81417)(0.65105)(20 \times 10^5) \text{ N/m}^2$$

$$P_r = 1.06 \text{ MPa}$$

$$\frac{A_3}{A_2} = \frac{A_4}{A_2} = 1.918 \Rightarrow \text{Mach} \text{ @ 2nd crit} \Rightarrow M = 0.32$$

M	P/P _t	T/T _t	A/A*	PA/P _t A*
0.32	0.93150	0.97993	1.92	1.79021

$$\Rightarrow P_3 = \left(\frac{P_3}{P_{t3}} \right) P_1 = (0.9315)(20 \times 10^5 \text{ N/m}^2)$$

$$P_3 = 1.863 \text{ MPa}$$

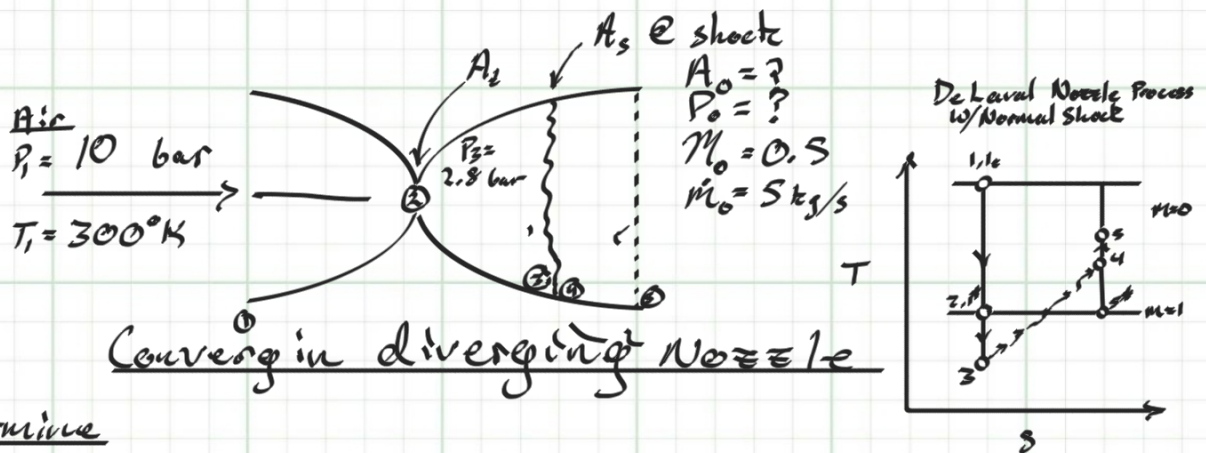
Results

$$A_4 = 19.2 \text{ cm}^2$$

$$P_r = 1.06 \text{ MPa @ 3rd crit.}$$

$$P_r = 1.863 \text{ MPa @ 2nd crit.}$$

Problem 2



Converging-diverging Nozzle

Determine

- a) A_4 b) A_3 c) P_0 d) A_0 e) The design Mach

Analysis

$$\frac{\dot{m}}{A^*} = \left(\frac{\gamma}{R} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}} \right)^{1/2} \frac{P_t}{\sqrt{T_t}} \rightarrow P_t = P_1, T_t = T_1$$

* $P_1 \approx P_{t1}$ for Nozzle no losses.

$$A^* = \frac{\dot{m} \sqrt{T_1}}{P_1 \left(\frac{\gamma}{R} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}} \right)^{1/2}} = \frac{\text{kg/s} \cdot (\text{K})^{1/2}}{\text{N/m}^2 \left(\frac{\text{kg} \cdot \text{m}}{\text{N} \cdot \text{s}^2} \cdot \frac{\text{K}}{\text{N} \cdot \text{m}} \right)^{1/2}} = \text{m}^2$$

$$A^* = \frac{(5)(300)^{1/2}}{10^6 \left(\frac{1.4}{2.87} \left(\frac{2}{2.4} \right)^{\frac{2.4}{0.4}} \right)^{1/2}} = A^* = 2.14265 \text{ cm}^2 = A_2$$

eg. 5.35 use Figure 5.5

$$P_{t1} A_1^* = P_{t2} A_2^* = P_{t3} A_3^* \Rightarrow \text{isentropic}$$

$$A_2 = A_1^* \quad P_{t1} = P_1 \rightarrow A_1^* P_{t1} = A_3^* P_{t3} = A_2^* P_{t2}$$

$$\therefore \frac{A_3 P_3}{A_2 P_{t1}} = \frac{A_3 P_3}{A_3^* P_{t3}} \Rightarrow \frac{A_4}{A_3^*} = 1 \quad A_4^* \neq A_3^*$$

Since losses take place across the shock

$$P_{t3} = P_{t4} \quad \& \quad P_{t1} = P_{t3} \rightarrow \frac{P_{t4}}{P_{t3}} = \frac{P_{t3}}{P_{t1}}$$

Since $M_3 = 0.5$ from Appendix G

M_3	P/P_t	T/T_t	A/A^*	P_A/P_{tA}	$\rightarrow$ isentropic table
0.5	0.84302	0.95238	1.33984	1.12931	

Since $M=1$ @ throat

M_1	M_2	P_2/P_1	T_2/T_1	$\Delta V/a_1$	P_{t2}/P_{t1}	P_{t2}/P_1
1.00	1.00	1.00	1.00	0	1.00	1.893

Using Appendix G1 - isentropic flow Parameters

M	P/P_t	T/T_t	A/A^*	$PA/P_t A^*$	$P_{t1} = P_{t2}$ No losses
1.00	0.528	0.833	1.00	0.528	

$$P_2 = \frac{P_2}{P_{t2}} P_{t2} = (0.528)(10 \text{ bar}) = 5.28 \text{ bar}$$

$$\frac{P_3}{P_{t3}} = \frac{2.8 \text{ bar}}{10 \text{ bar}} = 0.28 \Rightarrow M = 1.48 \Rightarrow \frac{A}{A^*} = 1.1629$$

$$\Rightarrow \left(\frac{A_3}{A_2} \right) (A_2) = A_3$$

$$A_3 = 24.9169 \text{ cm}^2$$

M_3	M_4	P_4/P_3	T_4/T_3	$\Delta V/a$	P_{t4}/P_{t3}	P_{t4}/P_3
1.48	0.70829	2.38880	1.30695	0.67077	0.936	3.33873

$$P_4 = \left(\frac{P_4}{P_3} \right) P_3 \Rightarrow P_4 = 6.68864 \text{ bar}$$

M_4	P/P_t	T/T_t	A/A^*	$PA/P_t A^*$
0.71	0.71448	0.90841	1.08729	0.77685

$$\Rightarrow \frac{P_t}{P_4} P_4 = P_{t4} \Rightarrow (0.71448)^{-1} (6.69) = 9.3615$$

$$\Rightarrow \frac{P_5}{P_{t5}} (P_{t4}) = (0.84302)(9.3615) = 7.892 \text{ bar} = P_0$$

$$A_3 = A_4 \Rightarrow \left(\frac{A_4}{A_4^*} \right)^{-1} A_3 = A_4^* \Rightarrow (1.08729)^{-1} (24.9169 \text{ cm}^2)$$

$$\left(\frac{A_5}{A_4^*} \right) A_4^* = (1.33984)(22.9165)$$

$$= 30.70 \text{ cm}^2 = A_5$$

$$\frac{A_5}{A^*} = \frac{30.70 \text{ cm}^2}{21.4265 \text{ cm}^2} = 1.4329 \Rightarrow M @ 3rd critical \approx 0.46906$$

or
 ≈ 1.80

Results

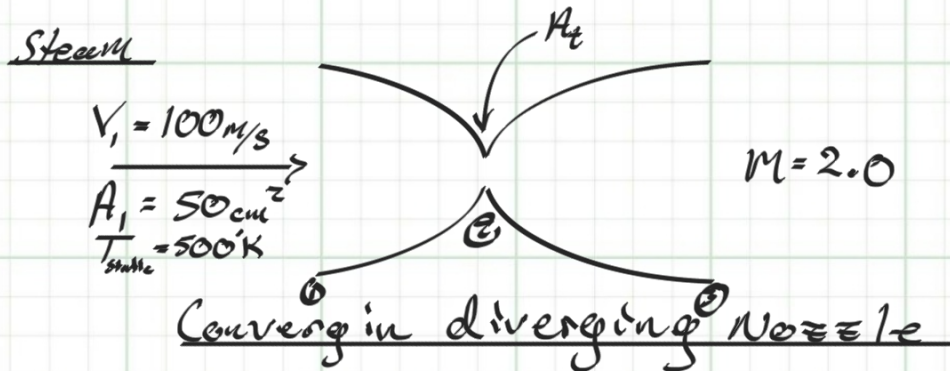
a) $A_{en} = A_2 = 21.4 \text{ cm}^2$

b) $A_5 = A_3 = 24.9 \text{ cm}^2$

c) $P_0 = 7.90 \text{ bar}$

d) $A_0 = A_5 = 30.7 \text{ cm}^2$

e) design Mach shock, $M_3 = 1.80$



Assume:

$$T_{\text{abs}} = T_{\text{amb}} + T_{\text{static}}$$

- isentropic flow
- Steam behaves as Perfect gas, w/ constant $\gamma = 1.3$

Determine: Throat and exit Areas.

$$M_1 = 1.0 \quad M_3 = 2.0 \quad T_{\text{abs}} = 273.15^\circ\text{K} + 500^\circ\text{K} = 773.15^\circ\text{K}$$

$$\frac{A_3}{A_2} = \frac{M_2}{M_3} \left(\frac{2 + (\gamma - 1) M_3^2}{2 + (\gamma - 1) M_2^2} \right)^{\frac{\gamma + 1}{2\gamma - 2}} \Rightarrow \frac{A_3}{A_2} = \frac{1}{2} \left(\frac{2 + (0.3) 2^2}{2 + (0.3)} \right)^{\frac{2.3}{0.6}}$$

$$\frac{A_3}{A_2} = 1.773 \quad A_2 = A^*$$

$$\frac{\text{kg m}}{\text{kg s}} \cdot \frac{\text{kg m}}{\text{kg s}} \cdot \text{K} \cdot \sqrt{\frac{\text{m}^2}{\text{s}^2}} = \frac{\text{m}}{\text{s}}$$

Need $M @ 1$ $M = \frac{V}{a}$, $a = \sqrt{\gamma g_c R T}$ $a = \sqrt{1.3 (461) (500)}$

Steam $M = 0.18 \Rightarrow \frac{A_1}{A^*} = 3.27793$

$$a = 547.4 \text{ m/s}$$

$$A^* = \frac{A^*}{A_1} A_1 = (3.27793)^{-1} (50 \text{ cm}^2)$$

$$M_1 = \frac{100 \text{ m/s}}{547.4 \text{ m/s}} = 0.18268$$

$$A^* = 15.25 \text{ cm}^2 = A_2$$

$$\Rightarrow A_3 = \frac{A_3}{A_2} A_2 = (1.773) (15.25 \text{ cm}^2) = 27.04 \text{ cm}^2$$

Results

$$A_2 = 15.3 \text{ cm}^2 \quad \Rightarrow \quad A_3 = 27.0 \text{ cm}^2$$

Area @ throat

Area @ Exit